How TimesFM-2.5 makes a forecast

Google · 231M parameters · decoder-only transformer · zero-shot

1. One series, read like text

The network reads one series at a time. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Known-ahead covariates do not enter the network. A linear fit on them (XReg) runs beside it, and TimesFM forecasts what the fit leaves over.

2. Scale as you go, then patch

The context is standardised once. Inside the model, each patch is scaled again with the mean and standard deviation of the values up to it.

$$\tilde{x}_k = \frac{x_k - \mu_{\le k}}{\sigma_{\le k}}$$

value

seen?

one patch = 32 steps

residual MLP

one token: 1,280 numbers

3. Twenty causal blocks

Each patch reads only itself and earlier patches. RMSNorm sits before and after every layer, with a shortcut around it.

earlier → later

the last patch reads them all

Causal self-attention: 16 heads, rotary positions (RoPE), normalised queries and keys

Feed forward: each token on its own, width 1,280, SiLU

4. 128 steps from the last token

The last token's output gives the next **128 steps**. A second head adds **9 quantiles** for up to 1,024 steps.

median

60% band

80% band

The scaling is undone and the median is the point forecast.

The library also forecasts −x and averages the two, and clips at zero a series that was never negative.

A causal model with 128-step outputs: one pass covers every horizon in the notebooks.

20 blocks · width 1,280 · 16 heads · patches of 32 in, 128 out · context up to 16,384 steps

In the notebooks: last 2,048 days or full monthly history · each metal on its own · XReg on D2's calendar covariates · no past-only covariates